

The Binomial Theorem

ANSWER KEY



Binomial coefficients

- 1 Evaluate each binomial coefficient:

a $\binom{6}{2} = 15$

b $\binom{8}{0} = 1$

c $\binom{5}{5} = 1$

d $\binom{10}{3} = 120$

- 2 Write down row 4 of Pascal's triangle (the row beginning 1, 4, ...) and explain how it relates to the expansion of $(a + b)^4$.

Row 4 is 1, 4, 6, 4, 1 — these are the coefficients of $a^4, a^3b, a^2b^2, ab^3, b^4$ in the expansion.

Expansions and specific terms

- 3 Expand $(x + 2)^4$ fully.

$$x^4 + 8x^3 + 24x^2 + 32x + 16.$$

- 4 Find the coefficient of x^3 in the expansion of $(2x - 1)^5$.

Term: $\binom{5}{2}(2x)^3(-1)^2 = 10 \cdot 8x^3 = 80x^3$. Coefficient: 80.

- 5 Find the term containing y^4 in the expansion of $(x + y)^7$.

$$\binom{7}{4}x^3y^4 = 35x^3y^4.$$

- 6 Use the binomial theorem to expand $(1 + 0.02)^5$ to the x^2 -style second-order term, and hence estimate 1.02^5 to four decimal places.

$$1 + 5(0.02) + 10(0.02)^2 + \dots = 1 + 0.1 + 0.004 + \dots \approx 1.1041 \text{ (true value } 1.10408\dots).$$

More expansions and terms

- 7 Expand $(2x - y)^3$ fully.

$$8x^3 - 12x^2y + 6xy^2 - y^3.$$

- 8 Find the middle term of the expansion of $(x + 3)^6$ (the term with x^3).

Middle term is the 4th term: $\binom{6}{3}x^3(3)^3 = 20 \cdot 27x^3 = 540x^3$.

- 9 Find the constant term in the expansion of $(x^2 + \frac{1}{x})^6$.

General term: $\binom{6}{k}(x^2)^{6-k}(\frac{1}{x})^k = \binom{6}{k}x^{12-3k}$. Constant term needs $12 - 3k = 0$, so $k = 4$: $\binom{6}{4} = 15$.

Sums of binomial coefficients

- 10 Evaluate $\binom{5}{0} + \binom{5}{1} + \binom{5}{2} + \binom{5}{3} + \binom{5}{4} + \binom{5}{5}$ without computing each term, using the identity $\sum_{k=0}^n \binom{n}{k} = 2^n$.
 $= 2^5 = 32$.
- 11 Use the fact that $\binom{n}{k} = \binom{n}{n-k}$ (symmetry of Pascal's triangle) to explain why $\binom{9}{2} = \binom{9}{7}$ without computing either value directly.
Choosing 2 items to include from 9 is equivalent to choosing 7 items to exclude, so the two counts must be equal — this is the symmetry property $\binom{n}{k} = \binom{n}{n-k}$ with $n = 9$.
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Further expansions

- 12 Find the coefficient of x^4y^3 in the expansion of $(x + 2y)^7$.
Term: $\binom{7}{3}x^4(2y)^3 = 35 \cdot 8x^4y^3 = 280x^4y^3$. Coefficient: 280.
- 13 Find the first three terms in the expansion of $(1 - 2x)^6$ in ascending powers of x .
 $\binom{6}{0}1^6 + \binom{6}{1}1^5(-2x) + \binom{6}{2}1^4(-2x)^2 + \dots = 1 - 12x + 60x^2 - \dots$.
- 14 Use the binomial theorem with $(1 - 0.01)^4$ to approximate 0.99^4 to four decimal places, using the first three terms.
 $1 + 4(-0.01) + 6(-0.01)^2 + \dots = 1 - 0.04 + 0.0006 = 0.9606$ (true value 0.96059601).